



OnePlus Buds Pro in Mithril

OnePlus Buds Pro gets all-new Mithril colour variant inspired by The Lord of the Rings. <https://digit.in/feb22-36>



WhatsApp's new feature soon

WhatsApp will soon allow users to listen to voice note even after closing chat window. <https://digit.in/feb22-37>

systems are also available with SoCs instead of CPUs. SoCs are designed for low-power applications and are way more economical than socketed CPUs. However, SoCs are only available with a motherboard since they are soldered and not socketed.

So, when you're trying to build a new PC, you need to chart out all the scenarios for which the PC will be used and mark out the outliers. This should give you a good idea about your compute requirements. And then you should pick the type of system you want. For example, if you wish to get a PC that's going to be used for data entry, then a thin-client system that uses an SoC should more than suffice. You'll have to add about 4GB of RAM along with a 500GB hard drive and install a lean operating system such

as Puppy Linux. If you wish to use Windows, then we suggest adding an additional 4GB of RAM.

Now, if you wish to make simple multimedia artwork for social media then we'd suggest that you use your smartphone. There are plenty of apps that can make quick work of your photos and produce brilliant artwork. However, if you wish to use a PC, then we're entering the Intel Core i3 or AMD Ryzen 3 territory, especially if you wish to edit locally using an open source software such as GIMP. However, if you happen to have access to an online editing service such as Canva which offloads all the processing workloads to their own servers, then you don't need more processing power locally. Now, if you're into video editing then you need to figure out the type of edits

that need to be done. Simple edits can be performed on your smartphones but for more elaborate yet simple edits, a PC with an Intel Core i5 or AMD Ryzen 5 processor would be ideal.

The cheapest way to get a complete PC would be to opt for one of the budget laptops which cost about 15-20K such as the ones made by RDP or Avita. These are completely standalone so there's little need to worry about configuring things. Those were all your options. There's always the refurbished market to check out but they come with added risks and there's no warranty coverage. If you've found another economical solution, then please do write in to us and let us know. We'll include it the next time we revisit this topic. [d](#)

BUYING ADVICE

SLOW NVME

Q Hi Agent001, I have a pretty powerful PC with the following configuration:

- CPU - AMD Ryzen 5 5600X
- CPU Cooler - CM Hyper 212
- MOBO - MSI B550M PRO VDH WIFI
- Graphics Card - RTX 3060
- RAM - 16 GB DDR4 3200 MHz
- SSD - Kingston KC2500 1TB
- HDD - WD Blue 2 TB
- PSU - Corsair CV650

I've noticed that my PC has become very sluggish over the last six months or so. It almost feels as if I don't have an SSD installed. My Windows partition is on the SSD and I also keep most of my applications on the same SSD. Only other files and backups are stored on the hard drive. But when I transfer files, my SSD is running super slow. I checked Task Manager also and it does not show high usage all the time.

—Bunty Singh

A Hey Bunty, would you happen to have installed way too many things on your SSD? There's a concept called provisioning that you need to do for your SSD. This is to ensure that your SSD lasts for the time that it is advertised for. The way an SSD works, it requires enough data to be available on hand to write an entire block. It



tries to avoid writing partial blocks as much as possible because of the limited read and write cycles that NAND chips have. Most SSDs leave out a little bit of space aside from the SSD firmware to juggle partial data blocks. This way, the usage fatigue is spread across the entire SSD and it helps prolong the life of the unit.

Things go for a toss when the amount of space that has been provisioned falls short of what is needed. We usually budget for anywhere be-

tween 10-to-25 per cent of the SSD's total capacity for provisioning. This varies based on the NAND technology, SSD firmware, controller and other factors. But a good rule of the thumb is to put aside 20-to-25 per cent of the storage space aside and never use that. This way, your NVMe or even SATA SSDs will continue to operate within the advertised limits.

In most cases, the advertised transfer speeds would be experienced up until you fill 15-to-20 per cent of the SSD's storage capacity. The transfer speeds would drop to about half after that. If you keep using up more space, let's say about 60-to-65 per cent, then your speeds will drop to about 20-to-30 per cent of the advertised speeds. This means, your speeds will drop from 2500 MBps to 1200 MBps and then down to 600 MBps as you keep using more and more of the SSD's capacity. At that point, your SSD will start feeling very sluggish. It'll still be way faster than a hard drive but if you want to get better speeds (or advertised speeds) then you need to leave plenty of unused space in your SSD. [d](#)